

Technical Occupational

Welding

Year 2 Learner`s Book

UNSAFE ACTS

- Employee behavior or actions that may risk safety in the workplace.



Chapter 1: Introduction (Orientation)

Chapter 2: Safety, First aid, HIV and AIDS (OHS Act)

Chapter 3: Measuring Equipment
Hand tools

Chapter 3: Materials

Chapter 4: Joining Methods

Practical Assessment Task: Projects

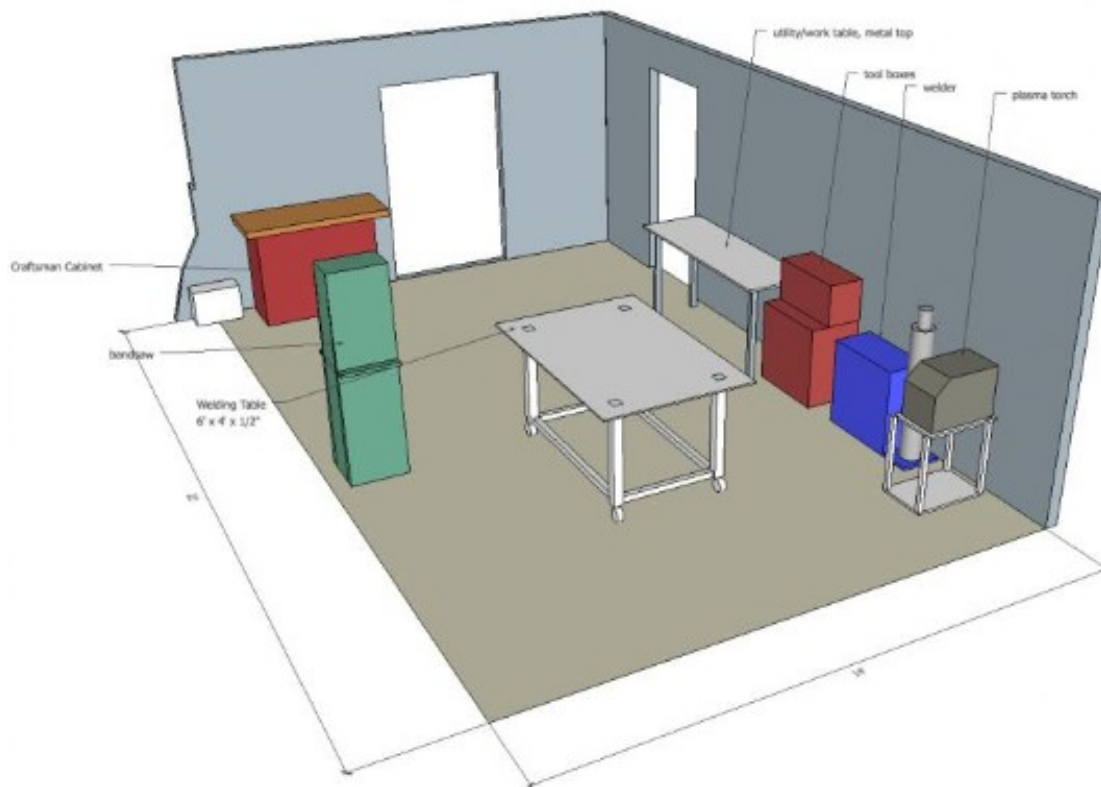
Glossary
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Orientation

Understand the Welding concept and the work place layout.

Welding is a fabrication process whereby two or more parts are fused together by means of heat, pressure or both forming a join as the parts cool. The resulting joint is known as a weldment and the conjoined parts are known as the parent material. The material used to create the weldment is called a filler or consumable.

Improving safety through layout and design



The layout of a welding workshop must be maintained and must be clear for anyone to see.

Passages/walkways must be clearly marked using yellow paint and nothing must block the walkways.

Machines and equipment must be arranged/designed in an orderly manner and this will help to avoid unnecessary accidents in the workshop.

Entering and exiting the workshop.

All exits and entrances are marked

You do not get in the workshop when the teacher is not around even if the entrance is open, wait for permission.

Observe the safety procedure by Walk softly when entering the workshop.

When exiting the workshop make sure that the tools you were using have been returned to their appropriate spots.

Clean up your work area and notify the workshop staff / teacher that you are leaving observing the safety procedure always.

- Take note of safety signs that indicates doors used for evacuation and entry.

Emergency signs always have a green background pasted on doors indicating exits and escape routes. They are in rectangular or square shapes with white pictogram on green background.



Emergency Assembly Point



Emergency Procedure

An emergency procedure is a plan of actions to be conducted in a certain order or manner, in response to a specific class of reasonably foreseeable emergency, a situation that poses an immediate risk to health, life, property, or the environment.

For example

When people are removed from hazardous location or situation which may affect to human healthy or life to safe place like assembly point.

Emergency evacuation procedure

- 1) Don't panic & stop the work.
- 2) Stop the equipment or running machines if possible.
- 3) For example if its fire then do the following:-
- 4) If the fire is small then try to attend to extinguish it by a suitably fire extinguisher otherwise leave it.
- 5) If the fire is big & it is not controlled then call / inform fire brigades or emergency response team.
- 6) Inform all workers by a whistle or calling.
- 7) Don't run but walk fast in a safe walkway.
- 8) Proceed to assembly point and don't re-enter.
- 9) Head count will start in assembly point to check if all learners have reached or any one is trapped / still in workshop

The safety evacuation drill

Fire drills are a vital part when it comes to your Fire Safety Plans.

It involves the evacuation of all the people within the building in an efficient way.

THE TYPE OF EVACUATION

Evacuation methods:

- A total building evacuation or single-stage.
- Evacuation from the dangerous area to a place where people can follow an escape route;
- Staff alarm which is an evacuation directed by staff (usually in shops or cinemas)
- Warning / alert signal evacuate. This type will allow for some time to investigate the potential source of fire followed by an evacuation signal if appropriate.
- Videos about safety

<https://youtu.be/DaYwcH1GMEg>

<https://youtu.be/WFm8BV1tYNs>

<https://youtu.be/ktrv34zW7-A>

Demonstrate in a class, in groups how to evacuate the workshop using a proper evacuation drill

Occupational Health and Safety

What is occupational health and safety?

Occupational health and safety (OHS) is a practice that deals with the safety, health, welfare and wellbeing of people when they are at work. Providing a safe working environment for staff is a legal responsibility for companies in South Africa and OHS requires both the prevention and treatment of any health issues that staff may experience during or because of their work.

Many mistakenly believe that accidents can never happen to them. As a result, work accidents often occur because people are careless and take chances. In fact, every year in South Africa, nearly 220 000 serious industrial accidents occur.

<https://youtube.com/watch?v=dBf6BTX1bmM>



What Causes Accidents - Safety Training Video - Preventing Accidents & Injuries.mp4

A definition

The Act defines an accident as an unplanned, uncontrolled event caused by unsafe activities and conditions

Causes of accidents

Accidents don't simply happen by chance: usually they result from carelessness.

The cause of accidents include:

- Poor housekeeping
- Loose clothing
- Improper use of tools
- Inaccurate setting-up of machines

GOOD HOUSE-KEEPING

WHAT IS HOUSEKEEPING AND EFFECTS OF POOR HOUSEKEEPING

What Is Housekeeping

Housekeeping means working tidily and orderly.

It also means general care, cleanliness, orderliness, and maintenance.



- Occupational health safety act (OSHA) regulations require that:
- We keep work environment clean.
- Always work neatly and return materials and tools to their proper places.

- Keep working areas unobstructed at all times; when using a machine, keep an unobstructed space (Of 2,25m²) around the machine.
- Grease, oil, tools, loose objects and materials lying around can easily cause a fall.
- Clearing job waste, clean up messes and spills
- Stacking materials, boxes and packages properly.
- A clean, orderly workshop is a safe workshop.

Effects of poor housekeeping

Here are some effect of poor housekeeping;

- Fire as a result of oily rags left in an area where hot work is performed, or due to the accumulation of combustible dust.
- Un-cleared waste could become fuel for the spread of fire.
- Littered job equipment, tools and materials could pose the risk of trip and fall.
- It can limit work space forcing workers to adopt poor working posture.
- Can maximize the severity /consequence of accidents.

Benefits of good housekeeping

1. It opens the door for deeper safety interactions:

2. It creates a sense of Order:

Good housekeeping is an indication that the rest of the workplace is well managed.

3. It exposes potential hidden hazards.

Hidden hazards get exposed when items are stacked and stored properly.

4. It improves productivity:

It reduces time that we spend looking for something that is lying behind a pile of rubbish.

5. It reduces slip and trip hazards:

- 6 It aids effective emergency response:

Escape routes are easier to identify in an emergency situation if proper housekeeping is in place.

It also creates a safe environment when having to get to the emergency exits.

- 7 It reduces fire hazards:

Many potential fire hazards are eliminated through proper housekeeping.

- 8 It allows for better identification of missing items:

If a workplace is well kept, tidy and having items/materials consolidated into distinct areas it makes it much easier to identify missing items/materials.

- 9 It makes stacked materials intrinsically safer:

- 10 It creates good first impressions:

Any new employee or learner will get off to a good start if they first enter a clean and orderly work site. Half the battle will be won.

Protective clothing

Personal protective equipment (PPE) is defined by Occupational Safety and Health Administration (OSHA) as any device or clothing that protects employees from injury or illness in the workplace.

This can include hard hats, safety glasses, earplugs, and safety shoes.



When welding you need to protect your eyes and respiratory system, as well as your body and your clothing.

You should protect yourself against heat, flying particles that are glowing with heat and radiation.

You should always:

Wear safety goggles when chipping, grinding or when drilling.

Not to wear loose clothing, jewellery, ornaments, a watch or key chains, or strap when working close to moving machinery.

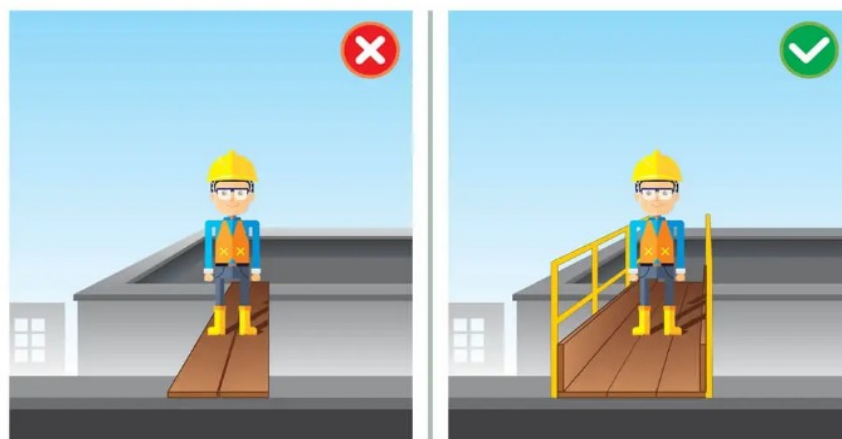
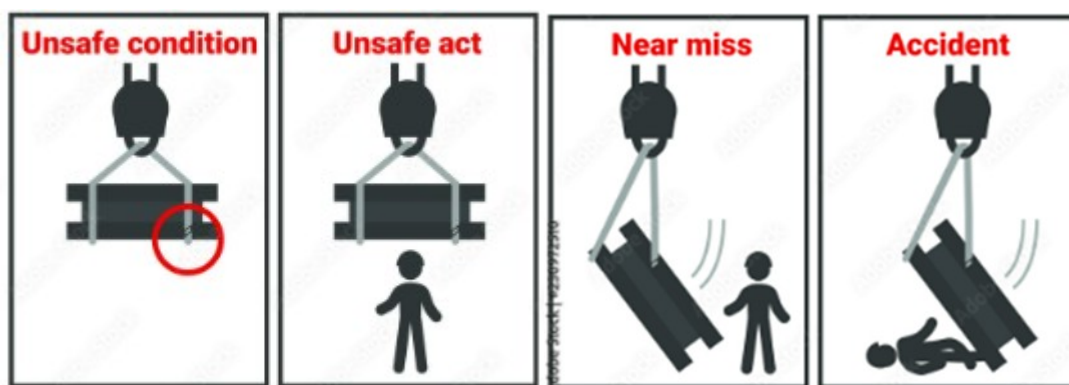
Different types of personal protective equipment.



UNSAFE ACTS & CONDITIONS

What are Unsafe Acts?

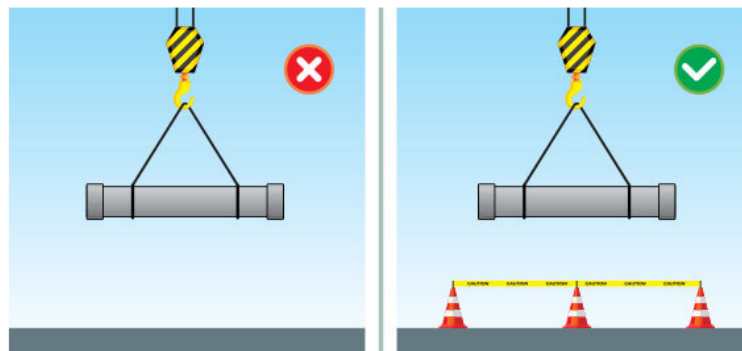
Unsafe acts refer to actions or behaviours that deviate from established safety protocols. These actions can range from disregarding safety guidelines to failing to use PPE correctly. Such acts not only jeopardize the individual involved but also pose risks to the entire workforce.

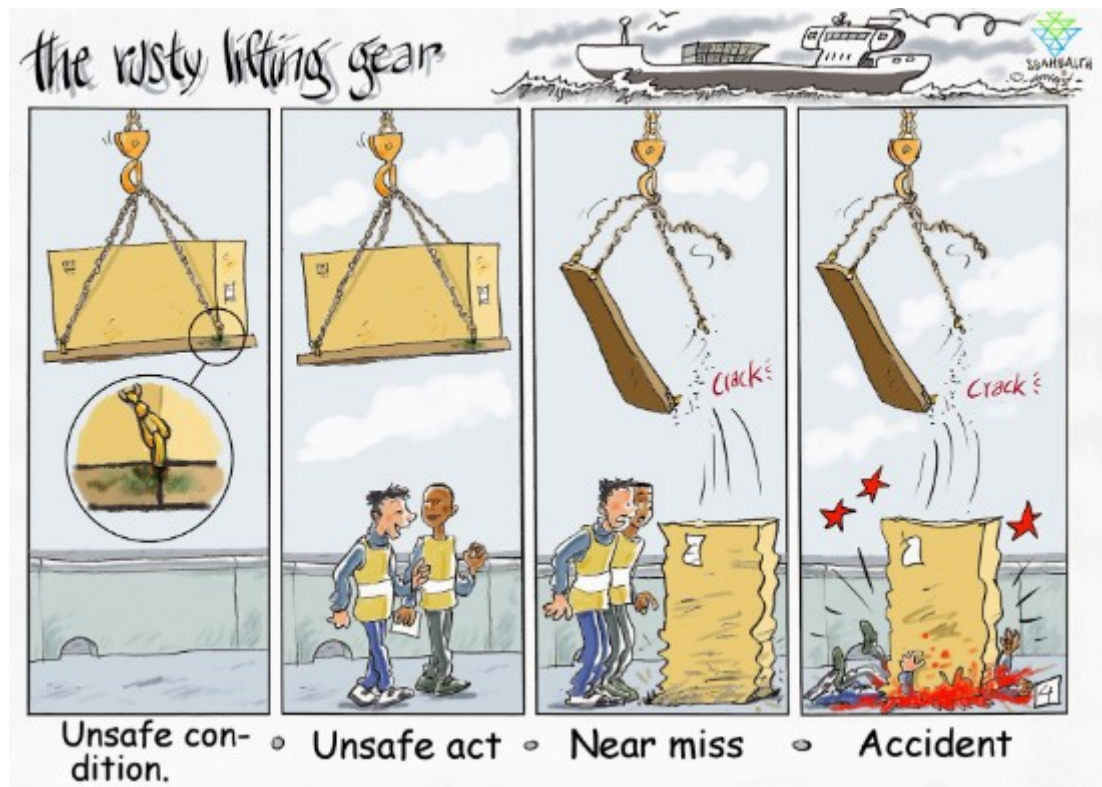




What are Unsafe Conditions?

Unsafe conditions encompass physical elements within the workplace that have the potential to cause harm. These may include poor housekeeping, inadequate lighting or ventilation, malfunctioning equipment, or improper handling of hazardous materials. Identifying and rectifying these conditions is essential to ensure the safety and well-being of employees.





Examples of common Unsafe Acts:

Some examples of unsafe acts commonly observed in workplaces include the following:

Not using personal protective equipment correctly or at all:

- Neglecting to wear safety goggles, helmets, gloves, or other necessary protective gear.
- Potential risks: Eye injuries, head trauma, burns, cuts, exposure to harmful substances.

Disregarding safety procedures and protocols:

- Ignoring established safety protocols, such as [lockout/tagout](#) procedures or confined space entry requirements.
- Potential risks: Equipment malfunction, electrocution, falls, asphyxiation, chemical exposure.

Engaging in horseplay or reckless behaviour:

- Engaging in rough or playful activities that distract or endanger oneself and others.
- Potential risks: Trips, falls, collisions, equipment damage, and severe injuries.

Operating machinery or equipment without proper training:

- Using machinery or equipment without adequate knowledge, training, or authorization.
- Potential risks: Crush injuries, amputations, entanglement, electrocution, equipment failure.

Ignoring ergonomic guidelines:

- Failing to follow ergonomic principles while performing tasks, such as lifting heavy objects incorrectly or maintaining poor posture.
- Potential risks: Musculoskeletal disorders, back injuries, strains, sprains, and chronic pain.

Examples of common Unsafe Conditions:

In addition to unsafe acts, there are various unsafe conditions that can compromise [workplace safety](#). It's crucial to identify and address these conditions promptly to create a safer work environment.

Poor housekeeping, leading to slips, trips, and falls:

- Cluttered workspaces, obstructed walkways, or spilled liquids can increase the risk of slips, trips, and falls.
- Potential hazards: Sprained or broken bones, head injuries, back injuries, and bruises.

Inadequate lighting or ventilation:

- Inadequate lighting can make it difficult to identify potential hazards or see warning signs, while poor ventilation can lead to air quality issues.
- Potential hazards: Increased likelihood of accidents due to reduced visibility, respiratory problems, eye strain, and fatigue.

Defective or malfunctioning equipment:

- Equipment that is not properly maintained or repaired can pose significant risks to employees.
- Potential hazards: Equipment malfunctions, electrical shocks, burns, and injuries caused by faulty machinery.

Improper storage or handling of hazardous materials:

- Failure to follow proper procedures for storing and handling hazardous materials can lead to accidents and exposure risks.
- Potential hazards: Chemical spills, fires, explosions, skin irritation, respiratory problems, and long-term health issues.

In our next section, we will learn about the potential consequences and why reporting and correcting these hazards is crucial.

Importance of Reporting and Correcting:

[Reporting unsafe acts and conditions](#) and taking prompt corrective measures are crucial for maintaining a safe work environment. Everyone in the organization, from management to employees, plays a vital role in ensuring workplace safety. Let's explore the importance of reporting and correcting unsafe acts and conditions:

Early Intervention:

- Reporting unsafe acts and conditions at the earliest opportunity allows for timely intervention and mitigation of potential risks.
- Identifying and addressing hazards promptly can prevent accidents, injuries, and illnesses before they occur.

Preventing Recurrence:

- Reporting unsafe acts and conditions helps identify underlying causes and patterns, enabling organizations to implement preventive measures.
- By addressing root causes, organizations can break the cycle of repeated [incidents](#) and create a safer work environment.

Accountability and Responsibility:

- Reporting unsafe acts and conditions fosters a culture of accountability and shared responsibility among all employees.
- It encourages individuals to take ownership of their own safety and the safety of their colleagues.

Continuous Improvement:

- Reporting unsafe acts and conditions provides valuable data for analyzing trends, identifying systemic issues, and improving safety protocols.
- Regular reporting and corrective actions lead to an ongoing cycle of improvement, ensuring a safer workplace over time.

Open Communication:

- Reporting unsafe acts and conditions encourages open communication between employees and management.
- It creates a supportive environment where employees feel comfortable raising concerns and suggestions for improving safety.

Unsafe Acts	Unsafe Conditions
• Operating equipment without authority • Failure to warn • Failure to secure • Operating at improper speed • Making safety device inoperable • Removing safety devices • Using defective equipment • Using equipment improperly • Failing to use personal protective equipment properly • Improper loading • Improper placement • Improper lifting • Improper position for task • Servicing equipment in operation • Horseplay • Under influence of alcohol and/or other drugs	• Inadequate guards or barriers • Inadequate or improper protective equipment • Defective tools, equipment or materials • Congestion or restricted action • Inadequate warning systems • Fire and explosion hazards • Poor housekeeping; disorderly workplace • Hazardous environmental conditions; gases, dusts, smokes, fumes, vapors • Noise exposures • Radiation exposures • High or low temperature exposures • Inadequate or excessive illumination • Inadequate ventilation

<https://youtu.be/2Jmo6Qy-1W4>

- **GENERAL WORKSHOP RULES**

Workshop rules and safety considerations

Workshop safety is everyone's responsibility, the following rules have been put in place to ensure the safety of all learners and staff. Please read the safety rules carefully before entering the workshop.

Workshop rules

- learners affected by drugs or alcohol are not permitted in the workshop

- learners with any health problems that may affect workplace safety (e.g., medication, epileptic fits) must report these conditions to the workshop staff
- Notify the workshop teacher of your arrival
- No food or drink in the workshop
- Wear the correct protective equipment for the tools you are using – ask if in doubt
- Immediately notify the workshop supervisor of any faulty or broken equipment
- Ask how to use the tools safely
- Make sure your work piece is fixed securely before work commences
- Keep clear of any person operating tools and machinery (bumping an operator or get tangled in the lead could cause serious injury to you or the operator)
- Do not talk to anyone operating electrical equipment and machinery
- Keep your work area tidy
- Clean up any spills immediately
- Wash hands after using equipment and materials

- Check that any tools you have been using have been put away in the appropriate spots, clean up your work area and notify the workshop staff.

<https://youtube.com/watch?v=-opN-c7hajY>



General Safety Rules.mp4

Clothing & footwear

- All loose clothing (e.g. shirts hanging out) must be tucked in.
- Safety boots or enclosed shoes must be worn in the workshop. Do not enter under any circumstances without this footwear, there are no exceptions to this rule.
- ***Remove rings and loose jewellery before entering the workshop they can be a hazard.***

Behaviour

fooling around and practical jokes in the workshop will not be tolerated.

First Aid

All accidents, cuts and abrasions must be reported before leaving the workshop. If an accident does happen, no matter how small, it must be reported to the workshop staff and an Accident Report Form filled out.

<https://youtube.com/watch?v=4bkr5lpKGUM>



Workplace Safety - Safety at Work - Tips on Workplace Safety.mp4

https://youtube.com/watch?v=u_keAFwiaQQ



Tony Crow, don't be blind to Safety.mp4

ACTIVITY 1

1. How can workers best contribute to positive safety culture in the workplace?
 - a) They should know how to complete work quickly if things go wrong.
 - b) They should report unsafe conditions after an accident has occurred.
 - c) They should report any unsafe conditions as soon as possible.
 - d) They should know the location of the first aid kit and accident book.
2. What does PPE stand for?
 - a) Protective preventative equipment
 - b) Personal protective equipment
 - c) Personal preventative equipment
 - d) People`s protective equipment
3. The legal responsibilities of an employer with regards to health and safety include:
 - a) Making employees pay for replacing damaged or lost PPE
 - b) Providing safe systems of work for all employees
 - c) Taking out additional insurances for dangerous work
 - d) Ensuring that only one member of the staff works on a dangerous jobs
4. Why would accidents in the workplace be reported?
 - a) To prevent similar accidents in future
 - b) To keep a record of it
 - c) To check whether staff are carrying out their duties correctly
 - d) To allow staff to learn from their mistakes
- 5 which of the following would prevent injuries from occurring at work?

- a) Keep the work place tidy
- b) Wearing light reflective clothing
- c) Lifting heavy objects with a bent back
- d) Keeping lighting to a minimum

6. If PPE is faulty, the work should:

- a) Not to do the task and inform the supervisor of the problem immediately
- b) Get somebody else to complete the task
- c) Use equipment and continue the task anyway
- d) Use the equipment to do the task and tell the supervisor latter

7. What is the best way of dealing with a hazard to ensure others are not put at risk?

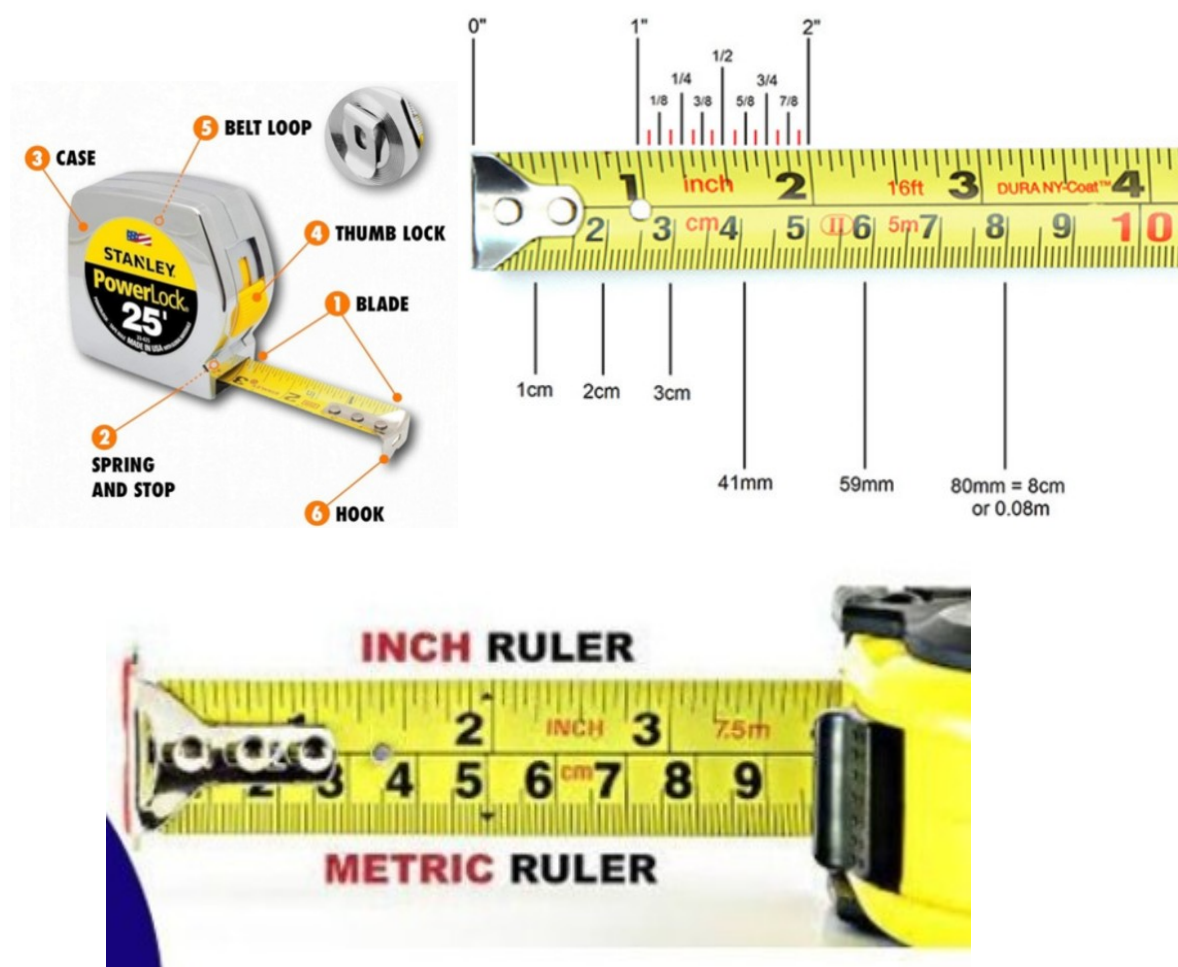
- a) Remove it immediately
- b) Leave it for the supervisor to sort it out
- c) Place a barrier tape around it
- d) Display a notice or warning sign

Select use & care: Hand measuring tools.

MEASURING TAPE

A tape measure is a type of hand tool used for measuring a wide range of materials.

It is to measure longer lengths; it is not as accurate as a steel rule. The tape is made of spring steel, cast steel or stainless steel

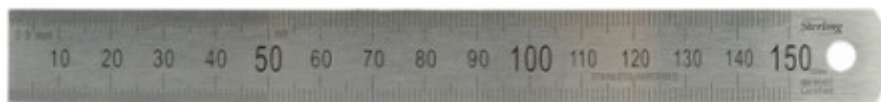


Case:

This part houses the rolled up tape measure and return spring.

Belt Loop:	A hook to place the measure tape on your belt when working on site
Spring and Stop	A spring loaded return and stop to help the tape measure back into the case.
Blade:	The spring steel tape with the markings on it you use to measure with.
Hook:	A bent piece of metal that helps you to hook the start of the measurement to the working side of the work piece.

- Steel ruler



A steel rule is a simple measuring instrument that is used for measuring distances and ruling straight lines. Steel rules are made of spring steel, cast steel, or stainless steel. It does not rust. Take care not to knock the edges.

- Meter stick



A meter stick is a large ruler used for measuring size or distance using the metric scale. The measurements provided on a meter stick are in centimeters (100 cm in a meter) and millimeters (1,000 mm in a meter).

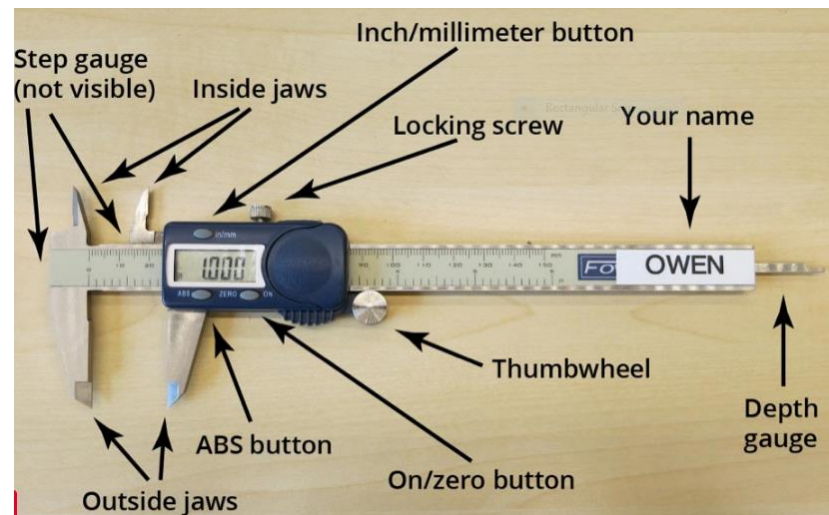
Activity

Name some uses of a steel tape

Name the materials steel rules are made of?

How should you look after a steel rule?

- Veneer Calliper



Vernier Calipers is for measuring the distance between two opposite sides of a surface. We can measure the internal and external dimensions and even the height of an object with accuracy. It can measure up to one-hundredth of a millimeter.

On/zero button: Turns on the calipers and sets the current position to zero

Thumbwheel: Opens and closes the jaws

Outside jaws: Measures the outside dimensions of an object

Inside jaws: Measures the inside dimensions of an object. These are delicate, so be careful.

Depth gauge: Measures the thickness of an object or the depth of a hole

Step gauge (not visible): Measures the distance from an edge to another point on an object

Locking screw: Prevents the caliper jaws from moving

Inch/millimeter button: Switches between inches and millimeters

ABS button: Temporarily sets the current position to zero

Battery compartment: Holds the battery

Your name: Tells people that these are your calipers. It's important in shared workspaces!

How to read the metric version of vernier calipers?

Regarding to the metric version of **vernier caliper**, there are three kinds of graduations: 0.1mm, 0.02mm, 0.05mm. How to read the metric version of vernier calipers correctly?

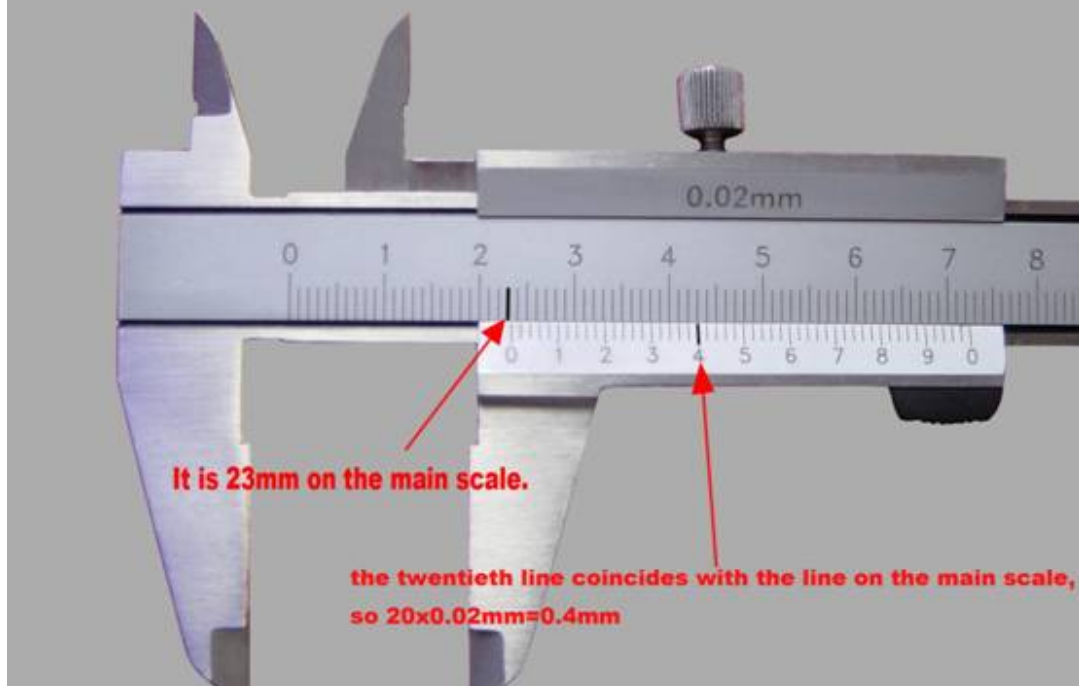
For the metric version of vernier caliper, each graduation on the main scale is 1.00mm. Every tenth graduation is numbered in sequence - 10 mm, 20 mm, 30 mm, 40 mm, etc. ...over the full range of the scale. It provides for direct reading in millimeters. The Vernier scale is graduated in 50 parts, each represents 0.02 mm. Every fifth line is numbered in sequence - 0.10 mm, 0.20 mm, 0.30 mm, 0.40 mm, 0.50 mm - providing for direct reading in hundredths of a millimeter.

Here we take the graduation of 0.02mm for example:

Three steps for reading the metric version of vernier caliper:

1. To count how many mm's line between the 0 line on the main scale and the 0 line on the Vernier scale. It is 23mm on the main scale.
2. Then to find the line on the vernier scale which exactly coincides with the line on the main scale. The twentieth line coincides with the line on the main scale, so $20 \times 0.02\text{mm} = 0.4\text{mm}$
3. Finally, add the reading in hundredths on the vernier scale to the reading on the main scale. $23\text{mm} + 0.4\text{mm} = 23.4\text{mm}$

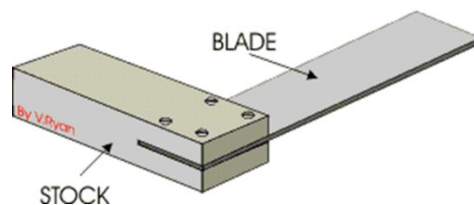
the final reading: $23\text{mm} + 0.4\text{mm} = 23.4\text{mm}$



An engineer's square is a tool used in metalworking and woodworking. It is mainly used to check if a workpiece is 'square', (that is, two sides are at 90 degrees to one another).

It can also be used to help draw or scribe a line at 90 degrees to the edge of a workpiece.

The stock is made of steel and the blade is hardened and tempered

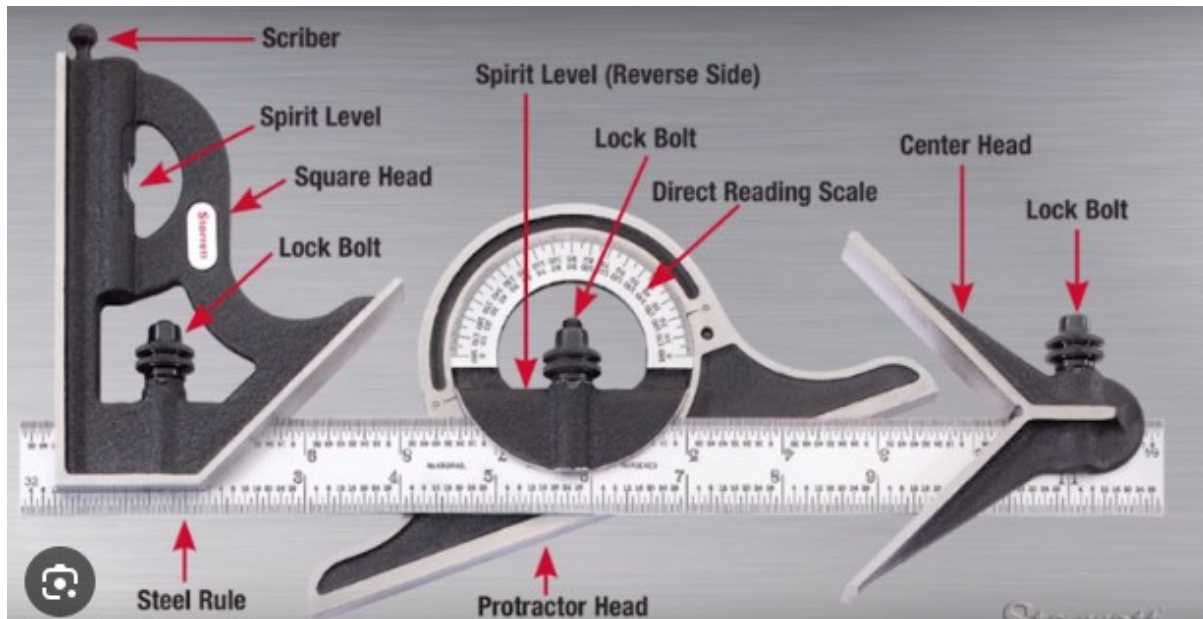


Stock:	The block part of the square that is sided up to the work side of the work piece.
Blade:	The blade of the square may have markings on it, but it is at a 90° angle with the stock.

Activity

1. What is an engineering square made of?
2. Explain briefly how to test a workpiece for squareness using an engineer's square.
3. Make a neat, labelled sketch of an engineer's square

Combination set



Combination sets consist of robust rule (with a groove cut along its centre) designed to take three different heads:

- a 90° – 45° square,
- a protractor
- a centre head.

The square head is used for:

- Checking and marking- off external and internal right angles
- As a depth gauge
- Checking and marking-off 45° angles
- The square head many be used alone as a spirit level

With the centre head fitted to the blade, the inside of the blade is exactly in the centre of the V formed by the legs. The centre head is used to determine the centre on the face of a round workpiece

Scriber and Marking gauge

A scriber is a hand tool used in metal work to mark lines on workpieces, prior to machining. The process of using a scriber is called scribing and is just part of the process of marking out.



A marking gauge is a device especially used for marking.



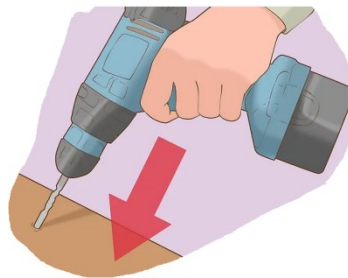
The proper procedure when drilling a hole into material.

Drilling Safely

- 1) Drill a pilot hole. In many cases, you'll get better results if you start with a drill bit a little smaller than the final hole size. Drill a shallow "pilot hole," then switch to the larger bit to finish the job. The pilot hole will help prevent your drill bit from slipping, and reduces the chance of splitting wood or other damage



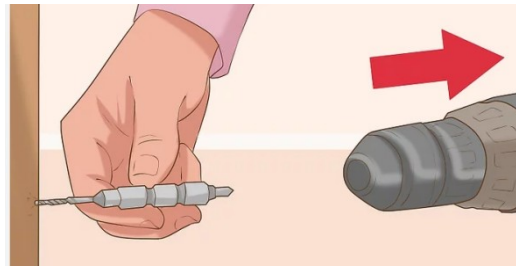
- 2) **Drill with steady pressure.** Hold the drill steady and push it into the material you're drilling. If it takes more than light force to drill the hole, you're probably using the wrong bit



- 3) Adjust the clutch. Each drill has a twistable collar to adjust the torque, often with a series of numbers on it. The higher the number, the more torque (rotational force) the drill will apply. If you are having trouble penetrating the material, increase the torque. If you are over-driving screws (burying them too deep), or if drilling too deep could damage something, lower the torque.
- 4) Avoid overheating the drill bit. If you are drilling through hard materials or drilling at high speeds, the drill bit will encounter an immense amount of friction. This can rapidly overheat the bit, to the point where it becomes red hot or burns the material you're drilling.



- 5) Do not force a jammed bit through. If the drill bit gets stuck in the material, don't try to force it out by running the drill. Unplug the drill, separate the bit and the chuck, and remove the bit using manual tools



The five-point step should be put into practice



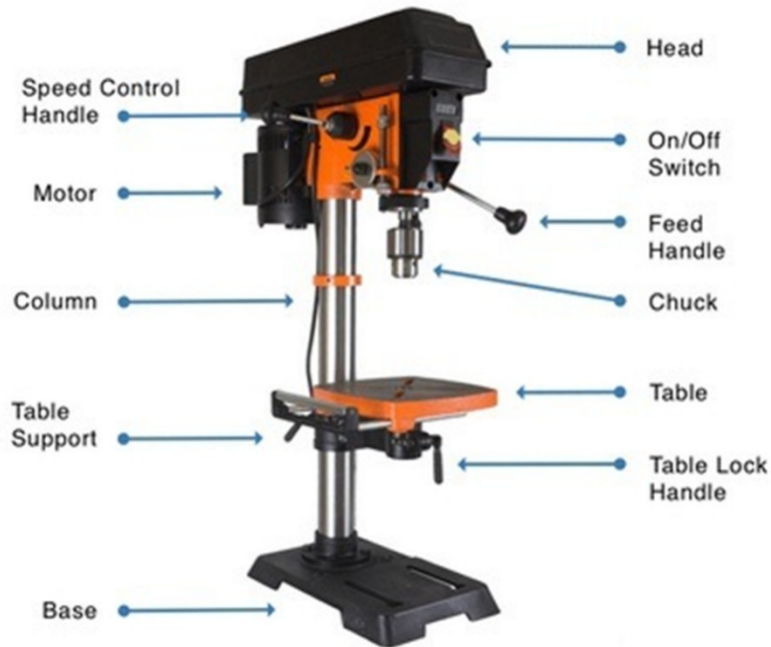
countersink tool

ELECTRICAL HAND DRILL



Strain Relief Cord:	A rubber grommet to help ease the strain on the mains cable.
Variable Speed trigger:	Allows you to manage the speed of the drill.
Trigger Lock:	For prolonged drilling to reduce stress on the hands, it keeps the drill on.
Reverse Function:	To reverse the direction of rotation.
Depth Stop	Depth Gauge allows you to stop at a certain depth when drilling many holes and all need to be the same depth
Safety Clutch and Clutch:	Determines the function of the drill. Hammer action or plain rotation.
Chuck:	A three jaw chuck for the fitting of drill bits.

Pillar drilling machine



Head:	Contains the drive pulleys and drive belt. It is covered for safety
On/Off Switch:	used to activate the drill.
Feed Handle	: Lowers and raises the chuck to control drilling depth.
Table:	Adjustable in height, the table is where the workpiece is placed.
Table Lock Handle:	Keeps the table stationary.
Base:	Gets fitted to the floor so the drill is stable.
Table Support:	Some drills have an extension on the table called a table support for larger pieces.
Column:	The backbone of the drill.
Motor:	The motor drives the pulleys that spin the drill bit.

Speed Control Handle:	Some drills adjust speed using the pulleys and some do so electronically

Chapter 3: Materials

Ferrous Metals



Non Ferrous metals_Types_Properties_Applications_Engineering materials_Aluminium_Zinc_Copper_GTU.mp4

The difference between ferrous metal and non-ferrous metal is that ferrous metals contain iron as their parent metal whereas non-ferrous metals are those metals which do not contain iron

Properties of ferrous metals are:

- Durable
- Great tensile strength
- Usually magnetic
- Low resistance to corrosion
- A silver-like colour
- Recyclable
- Good conductors of electricity

Low Carbon Steel (Mild Steel)

Low carbon is commonly known as mild, soft or machinery steel. It contains 0.15...0.3% of carbon. They are pretty low-cost and (malleability) very well-suited for bending operations. The surface hardness can be increased through carburising.

Some examples include bolts and nuts, forgings, medium-loaded details, etc.

Medium Carbon Steel

Medium carbon steels contain 0.3%...0.75% of carbon. The higher carbon content increases their strength and hardness compared to low-carbon

steels. At the same time, the ductility decreases. The increase of carbon and manganese allows tempering and quenching.

Medium carbon steels are mainly used for making different automotive industry components like gears, axles, and shafts but also bolts, nuts, screws, etc. Steels ranging from 0.4...0.6% are also suitable for everything related to locomotives and rails.

High Carbon Steel

High carbon steel is also known as tool steel, with a carbon content ranging between 0,75% to 1,7%. Used for most cutting tools, also used for making chisels, files, drills, hammers, taps and crowbars.

Cast iron is made from pig iron, which is the product of melting iron ore in a blast furnace. Cast iron can be made directly from the molten pig iron or by re-melting pig iron, often along with substantial quantities of iron, steel, limestone, carbon (coke) and taking various steps to remove undesirable contaminants.

NON FERROUS METAL



Non Ferrous metals_Types_Properties_Applications_Engineering materials_Aluminium_Zinc_Copper_GTU.mp4

The two metals which comes under Non-Ferrous metals are

1. Aluminium
2. Copper

Aluminium:

- Aluminium is a lightweight metal with good formability.
- M.P of Al is 658 °C
- The pure metal is soft and weak but when employed with copper, magnesium and silicon or other metals, it becomes harder and stronger and is employed for various applications.

Properties of Aluminium:

High electrical & thermal conductivity.

Aluminum is non-magnetic and has the property of corrosion Resistance.

Applications of Aluminium:

- It is a good conductor of electricity & cheap so it is used for making electrical wires.
- As it is ductile in nature, it can be used to protect pure metals by Aluminium foil.
- It is used in furniture, railroad, trolley cars, automobile bodies and pistons, electric cables, rivets.

Aluminium Alloys:

Aluminium alloys are alloys in which aluminium (Al) is the most important metal. The typical alloying elements are magnesium, copper, manganese, silicon, tin and zinc.

Some of the important alloys of Aluminium are

1. Duralumin (copper, aluminum)
2. Commercially pure aluminium

3. Y-Alloy

Applications:

Used for automobile pistons and other aircraft parts.

Copper is reddish brown in colour and is tough, ductile and malleable.

Copper is a pure metal it bends and stretches without fracture

- It has good electrical conductivity
- Corrosion resistance
- Good thermal conductivity

Applications of Copper:

- For making electrical cables.
- Heat exchangers
- For making various copper alloys, such as bronze and brass.
- For household appliances, bearings, etc.

Non- Ferrous Alloys of Copper are as follows.

Brass:

- Brass is an alloy of mainly zinc and copper.
- Brass contain zinc as the principal alloying element

- In various brasses, amount of zinc varies from 5 to 40% and that results in increasing strength and hardness.

Bronze:

- It is basically an alloy of Copper and Tin. Bronze is usually harder than brass.
- It possesses superior mechanical properties and corrosion resistance than the brass.
- It can be rolled into wire, rod, and sheets.

Types of Bronze:

The metals which comes under Bronze are

1. Phosphor Bronze
2. Aluminium Bronze
3. Coin Bronze

Bronze is used for valves, valves seats, bearings and gear wheels.

Advantages of Non-Ferrous Metals:

- Easy to fabricate, i.e., machining, casting, welding, forging and rolling.
- High corrosion resistance and low density.
- Possess very good thermal and electrical conductivity.

Identification of metals

How to use a magnet to show the difference between non-ferrous and ferrous materials.

The spark test is by using a bench grinder shown spark patterns differ with different materials (backup test for non & ferrous materials)

Methods that can be used to identify Metals in the workshops are:-

Spark test

High- carbon steel: short, very white or light yellow carrier lines with considerable forking with many star-like bursts

<https://youtube.com/watch?v=GnSBSKTC834>

Low-carbon steel: straight carrier lines with a yellowish colour with a small amount of branching and very little carbon burst

Cast iron: short carrier lines with many bursts, which are red near the grinder and orange-yellow farther out; considerable pressure is required on cast iron to make sparks.

After completing this chapter, you will be able to:

Produce semi-permanent joints using any two of the following methods:

Rivets, bolts and nuts, studs and split pins on a flat bar or sheet

What are semi-permanent joining applications?

Semi-permanent joint application refer to methods used to join different objects together, securely but not necessarily not permanently.

The methods are as follows:-

Rivets

Bolts and nuts

Studs

Locking devices

Split pins and keys

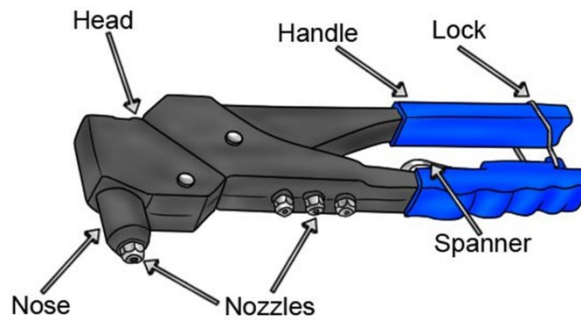
Rivets

Rivets are divided into two rivets: blind rivet / pop rivets and solid rivets

Mechanical technology: fitting and machining by D Meyer BH Van der Westhuizen

Grade 10 pages95-112

JOINING TOOL : RIVET GUN



To use a riveter you need to have the correct sized nozzle to fit the rivets you are installing.

Open the handles of the riveter and place the mandrel of the rivet into the nozzle.

The mandrel is the long shaft that is slotted through the body of the rivet.

It is slotted into the riveter when the rivet is being fastened. The riveter pulls the mandrel through the body of the rivet, expanding the pin and then snapping off the mandrel.

Slot the body of the rivet through the hole that has been drilled in the material that you are fastening.

Push the riveter against the material gently to make sure the rivet is fully inserted.

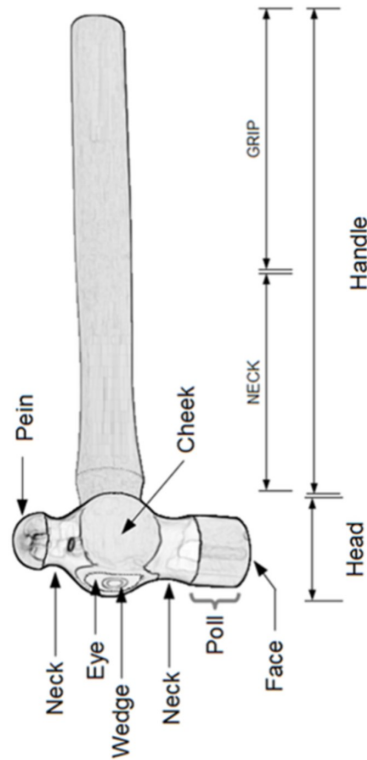
Squeeze the handles with one or two hands depending on the type of riveter you are using

Bring the handles as close together as you can when squeezing. This will create a second head to secure the rivet in place and pull the waste part of the mandrel out of the rivet

<https://youtube.com/watch?v=7M0RsmR5d6M>



BALL PEIN HAMMER



Pein:

The rounded end of the hammer used to tap down and shape metal brims and other metal parts you want to shape

Neck:

The shaped part leading up to the pein and to the poll.

Eye:

A hole in the hammer head through which the handle is inserted

Wedge:

A wedge shaped insert that splits open the handle tip in order to firmly seat it in the head.

Poll:

The thickened tool steel portion which give the hammer its head size and weight.

Head:

The striking end of the hammer which should be honed from time to time to keep it true and effective.

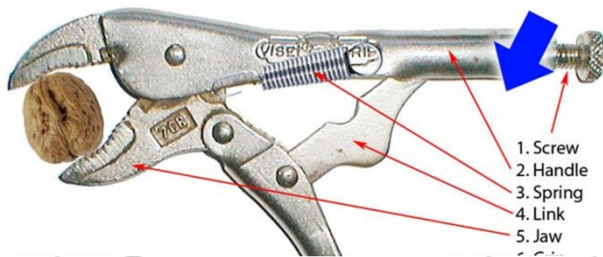
Cheek:

The centre part of the hammer through which the neck is forged.

Handle:

Made from hickory wood it is hard and tough to withstand the shock of being hammered constantly. The grip is the part which is held in the dominant

VICE GRIP



Screw:

The screw is used to adjust the jaw opening

Handle/s:

The handles are used to close and lock the vise-grip into position.

Spring:

This helps the jaws open when the release lever is activated

Link:

The link is the locking mechanism of the vise-grip.

Grip:

The grip is manoeuvred with the opposing fingers of the hand to press

PUNCH

A dot punch is used to mark off a distance without making a large impression	
A centre punch is meant to make a larger impression so a drill can be centred thereon to drill a hole in the metal	
Anvil:	Back of the punch that is struck with the ball peen hammer.
Handle:	The knurled area used to hold on to the punch firmly.
Tip:	The working end of the punch that makes the impression on the work piece. The tip needs to be honed from time to time on the bench grinder.

ANGLE GRINDER



Power Cable:	Cable used to connect to mains.
Inlet:	Cool air is sucked in here to cool the motor down.
Removable Vice Handle	An additional handle to help control the angle grinder.
Safety Mask:	Protects the user against excessive sparks and disk breakage.
Cutting Disk:	This is the replaceable disk that is used up during the use of an angle grinder.
Metal Head:	Contains the gearbox of the grinder.
Spindle Lock:	Locks the axle of the gearbox so the cutting disk can be changed.
Air Outlet:	Hot air is vented here as the motor and gearbox is cooled down.
External Carbon Brush:	When the brushes are used up, it is replaced at the receptacles.

Welding

Welding is a fabrication process whereby two or more parts are fused together by means of heat, pressure or both forming a join as the parts cool.

Welding is usually used on metals and thermoplastics but can also be used on wood. The completed welded joint may be referred to as a weldment.

The parts that are joined are known as a **parent material**.

The material added to help form the join is called **filler** or **consumable**.

The form of these materials may see them referred to as parent plate or pipe, filler wire, consumable electrode.

<https://youtube.com/watch?v=QsayOd32i2E>



Stick Welding_#1 Beginner Mistake and How to Fix It.mp4

Consumables are usually chosen to be similar in composition to the parent material, thus forming a homogenous weld,

The completed welded joint may be referred to as a **weldment**.

How Does Welding Work?

Joining Metals

welding is a high heat process which melts the base material. Typically with the addition of a filler material.

Heat at a high temperature causes a weld pool of molten material which cools to form the join, which can be stronger than the parent metal. Pressure can also be used to produce a weld, either alongside the heat or by itself.

It can also use a shielding gas to protect the melted and filler metals from becoming contaminated or oxidised.

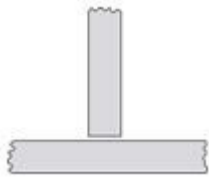
Common Joint Configurations

Butt Joint



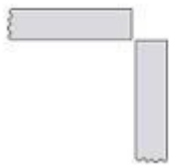
A connection between the ends or edges of two parts making an angle to one another of $135\text{-}180^\circ$ inclusive in the region of the joint.

T Joint



A connection between the end or edge of one part and the face of the other part, the parts making an angle to one another of more than 5° up to and including 90° in the region of the joint.

Corner Joint



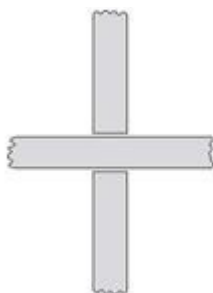
A connection between the ends or edges of two parts making an angle to one another of more than 30° but less than 135° in the region of the joint.

Edge Joint



A connection between the edges of two parts making an angle to one another of 0 to 30° inclusive in the region of the joint.

Cruciform Joint



A connection in which two flat plates or two bars are welded to another flat plate at right angles and on the same axis.

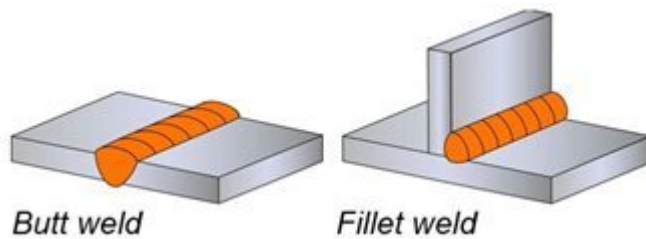
Lap Joint



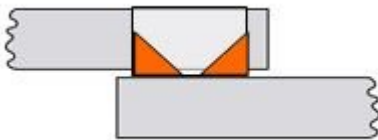
A connection between two overlapping parts making an angle to one another of 0-5° inclusive in the region of the weld or welds.

Types of Welding Joints

Welds Based on Configuration

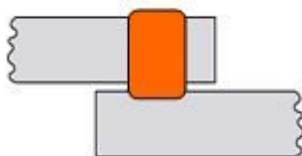


Slot weld



Joint between two overlapping components made by depositing a fillet weld around the periphery of a hole in one component so as to join it to the surface of the other component exposed through the hole.

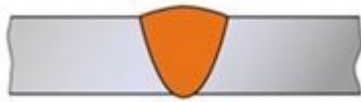
Plug weld



Weld made by filling a hole in one component of a workpiece with filler metal so as to join it to the surface of an overlapping component exposed through the hole (the hole can be circular or oval).

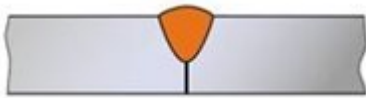
Based on Penetration

Full penetration weld



Welded joint where the weld metal fully penetrates the joint with complete root fusion

Partial penetration weld



Weld in which the fusion penetration is intentionally less than full penetration.

Welds Based on Accessibility

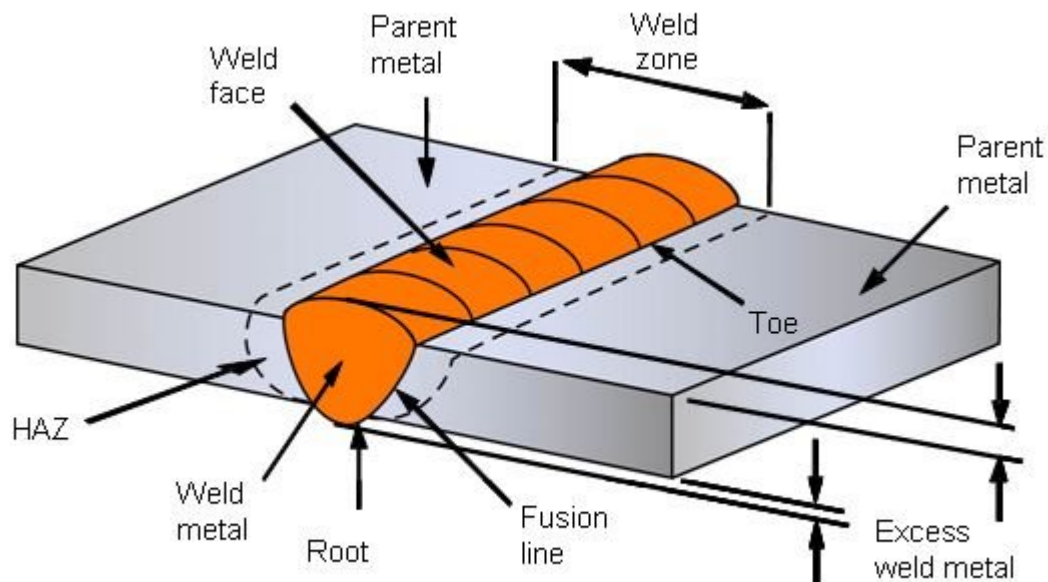


Single side weld

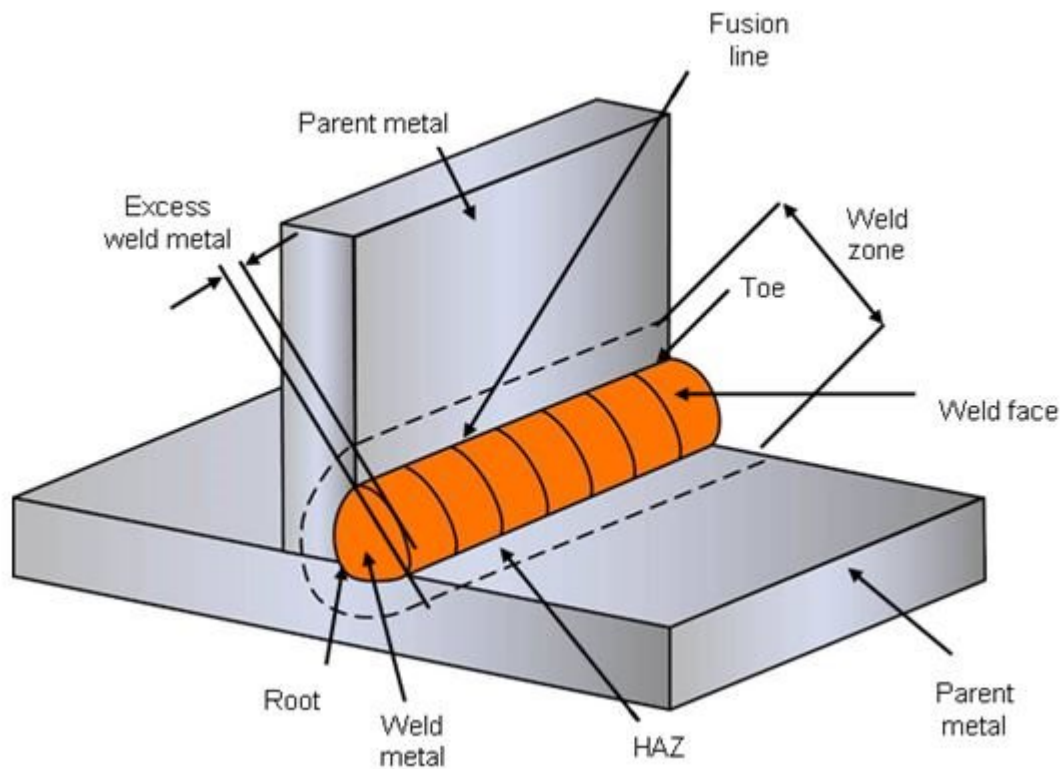


Double side weld

Features of Completed Welds



Butt weld



Fillet weld

Parent Metal

Metal to be joined or surfaced by welding, braze welding or brazing.

Filler Metal

Metal added during welding, braze welding, brazing or surfacing.

Weld Metal

All metal melted during the making of a weld and retained in the weld.

Heat Affected Zone (HAZ)

The part of the parent metal metallurgically affected by the weld or thermal cutting heat, but not melted.

Fusion Line

Boundary between the weld metal and the HAZ in a fusion weld. This is a non-standard term for weld junction.

Weld Zone

Zone containing the weld metal and the HAZ.

Weld Face

The surface of a fusion weld exposed on the side from which the weld has been made.

Weld Root

Zone on the side of the first run furthest from the welder.

Weld Toe

Boundary between a weld face and the parent metal or between runs. This is a very important feature of a weld since toes are points of high stress concentration and often they are initiation points for different types of cracks (eg fatigue cracks, cold cracks).

In order to reduce the stress concentration, toes must blend smoothly into the parent metal surface.

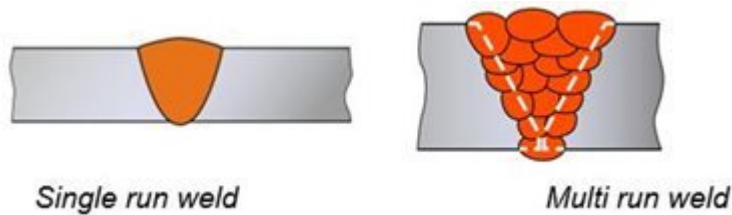
Excess Weld Metal

Weld metal lying outside the plane joining the toes. Other non-standard terms for this feature: reinforcement, overfill.

Note: the term reinforcement, although commonly used, is inappropriate because any excess weld metal over and above the surface of the parent metal does not make the joint stronger.

In fact, the thickness considered when designing a welded component is the design throat thickness, which does not include the excess weld metal.

Run (pass)



The metal melted or deposited during one passage of an electrode, torch or blowpipe.

Layer

Stratum of weld metal consisting of one or more runs.

Energy Sources

Different processes are determined by the energy source used, with a variety of different techniques available.

What are the Different Types of Welding and What are They Used for?

There are a variety of different welding process types with their own techniques and applications for industry, these include:

1. Arc

This category includes a number of common manual, semi-automatic and automatic processes. These include [metal inert gas \(MIG\) welding](#), stick welding, [tungsten inert gas \(TIG\) welding](#) also known as gas tungsten arc welding (GTAW), gas welding, [metal active gas \(MAG\) welding](#), [flux cored arc welding \(FCAW\)](#), gas metal arc welding (GMAW), submerged arc welding (SAW), shielded metal arc welding (SMAW) and plasma arc welding.

These techniques usually use a filler material and are primarily used for joining metals including stainless steel, aluminium, nickel and copper alloys, cobalt and titanium. [Arc welding](#) processes are widely used across industries such as [oil and gas](#), [power](#), [aerospace](#), [automotive](#), and more.

2. Friction

3. Electron Beam

4. Laser

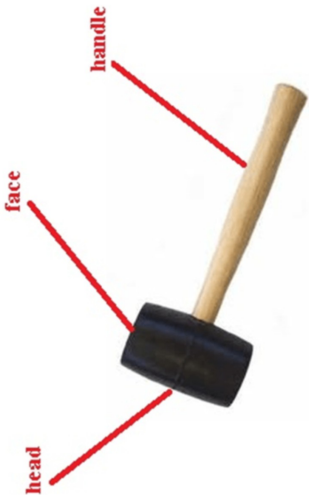
5. Resistance

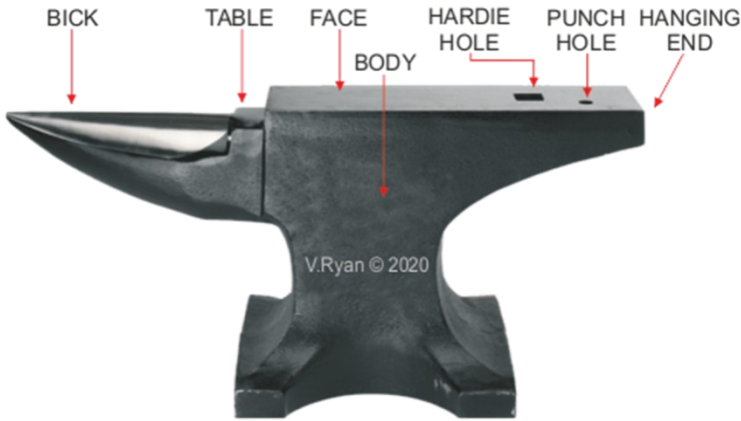
[Spot welding](#) uses heat delivered between two electrodes which is applied to a small area as the workpieces are clamped together.

	Fulcrum:	This is the pivot point around which the parts of the pliers rotate.
	Jaws:	The jaws has a specific size and determines the size of the pliers.
	Pipe Grip	The rounded shape of the pipe grip allows you to hold on tight to rounded objects, such as pipes.
	Tip	The tip is used to pinch or hold an object with precision

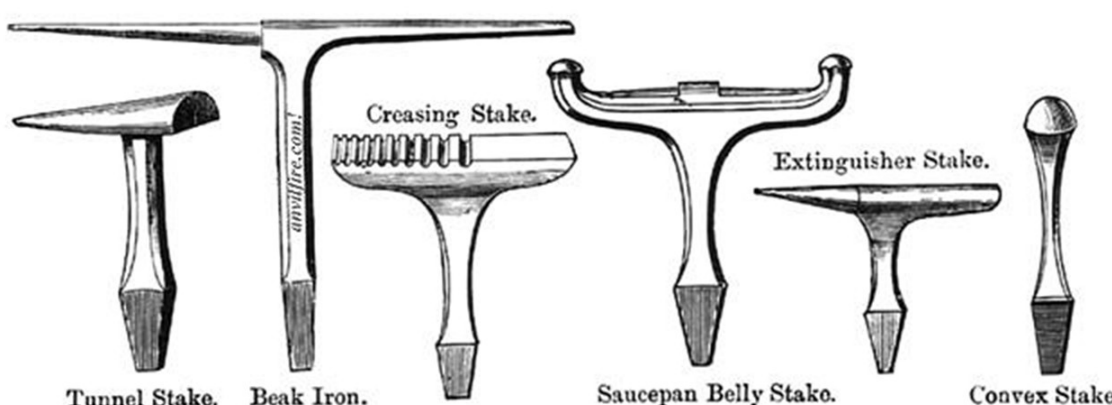
SIDE CUTTER		
 The diagram shows a side cutter with red and black handles. Labels with arrows point to the 'Joint' (the pivot point), 'Handles' (the long grips), 'Jaws' (the curved ends used for gripping), and 'Blade' (the sharp edges used for cutting).	Handles:	The handles are used to grip the side cutter. Some side cutters have insulated handles to cut electrical wires. Never cut electrical wires while the power is on!
	Jaws:	The jaws determine the size of wire that can be cut.
	Blade:	The blade is the working end of the side cutter. Never cut wire or objects that are too hard for the blades to handle. Generally you should only cut copper wire with side cutters.
	Joint/ Fulcrum:	This is the pivot point around which the parts of the pliers rotate.

MALLET HAMMER

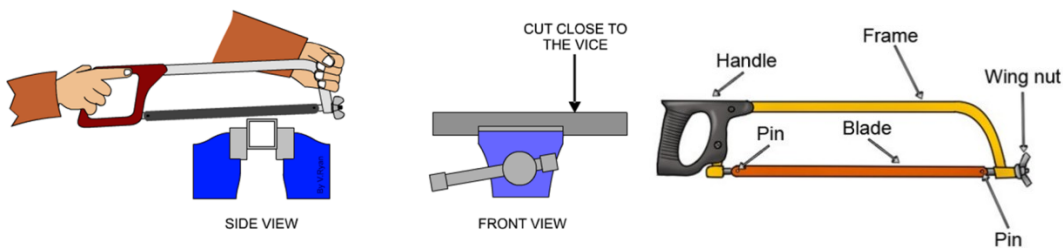
	Handle:	The part held onto when applying a blow.
	Face:	The hammer head face makes contact with the workpiece.
	Head:	The head of the mallet can be made from various firm materials such as plastic, rubber, copper and traditionally wood

THE ANVIL	
	
Bick/Horn	It is shaped like a horn and is used to forge rounded shapes over
Table:	A slightly rounded flat area used to give a slight contour to the item being forged.
Face:	The large flat area where most of the forging/hammering takes place,
Punch Hole:	This is a round hole used in punching holes through the entire metal that you are working on

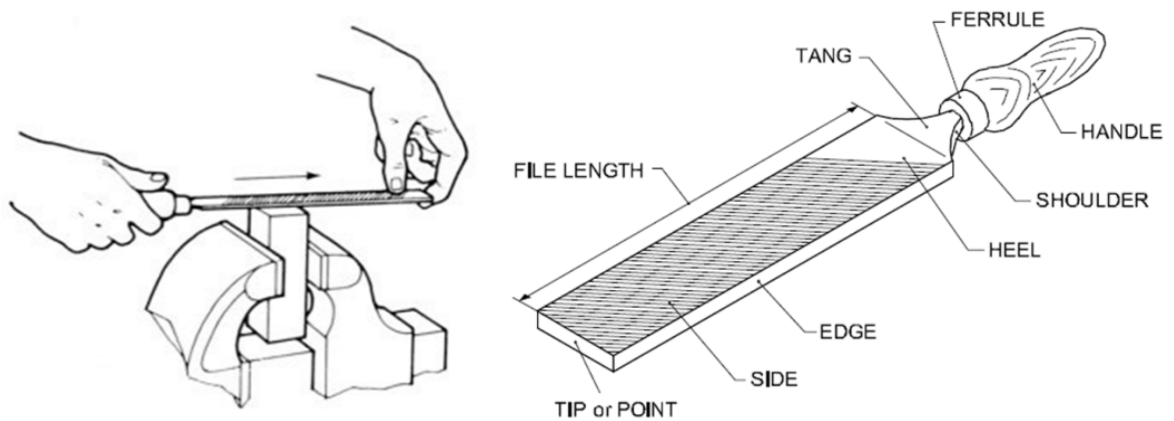
Hanging end	This is an overhanging part which allows the artisan to forge a piece that needs to fit around a corner or have a unique shape.
Hardie Hole:	The stakes are positioned into the hardie hole when doing specialist work on the stake. It can also be used to forge a square hole through a workpiece.
Body:	Made of solid wrought or cast iron, this area gives the anvil its weight, making it a stable work surface.

THE STAKES	
	
Tunnel Stake	Forms funnels and taper shapes
Convex Stake:	For shaping domes and rounded hollow shapes.
Extinguisher stake:	For bending and trueing a piece – making it to an exact shape.
Saucepan Belly Stake	To shape rounded shapes into sheet metal.

Creasing Stake:	Used with a rounded die or peen hammer it is used to form grooves and was used before sheet metal rollers were invented.
Beak Iron:	To forge a spout or small long tube

HACKSAW	
 The image contains three diagrams of a hacksaw. On the left, a 'SIDE VIEW' shows a person's hands using the saw to cut a piece of metal held in a blue vise. In the center, a 'FRONT VIEW' shows the saw blade positioned to cut a metal bar, with an arrow pointing to the cut line labeled 'CUT CLOSE TO THE VICE'. On the right, an exploded view diagram labels the components: 'Handle' (black), 'Frame' (yellow), 'Blade' (orange), 'Pin' (two locations), and 'Wing nut' (at the end of the frame).	
Pins	The pins hold the hack saw blade in place.
Blade:	Made from strong steel blades come in different sizes or teeth per inch.
Handle:	This is shaped to fit your hand and you will take this part in your dominant hand.
Frame:	The extension of the handle to help the blade's full length fit to the saw.
Wing Nut:	The wing nut tightens the blade and should be relaxed between uses to the blade does not stretch out or the frame becomes buckled. The wing nut is important because the blade needs to be taut.

FILES



Clamp the work piece securely in a bench vice.

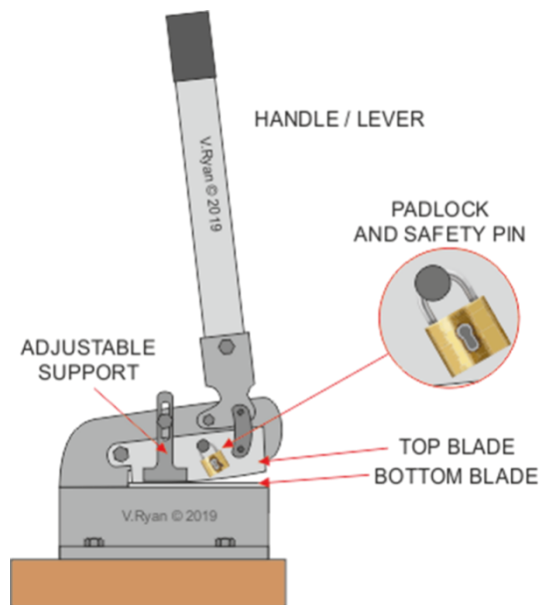
Ensure that you have scribed the target line clearly onto the work surface on the operator side.

Stand squarely in front of the workpiece with your feet slightly apart, the dominant foot slightly in front of the other.

Using your dominant hand on the handle and the other hand on the tip to steer the file, press down slightly and file away from your body.

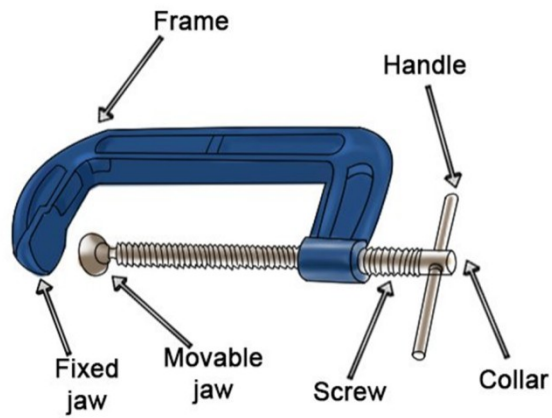
Slightly lift the file, retract and repeat. Make sure you inspect the cutting edge of the file and the target line regularly and adapt your file and body position to correct the cut.

HAND GUILLOTINE/SHEAR

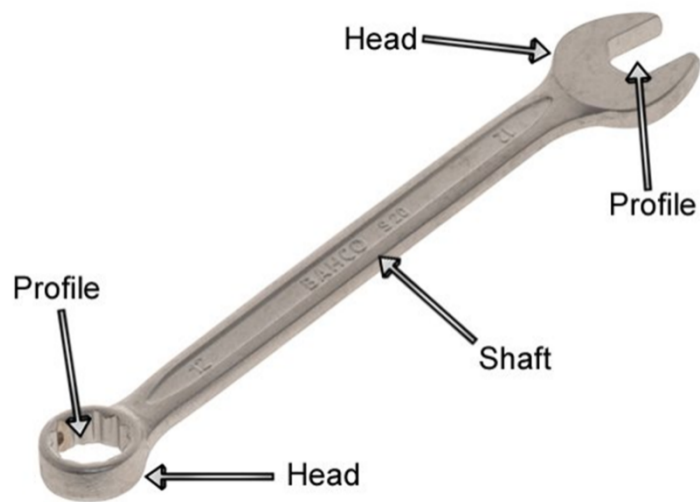


Adjustable support:	Adjusted to the thickness of the material to keep it from flying up during a cut.
Top and Bottom Blade:	Never cut round bars with the top and bottom blade, it will notch the blade and spoil the cut
Padlock and Safety Pin:	A safety device to prevent accidents.
The handles are quite long, look behind you before you cut and ensure nobody is behind you that can get hurt.	
Only cut the thickness material that the shear can handle.	

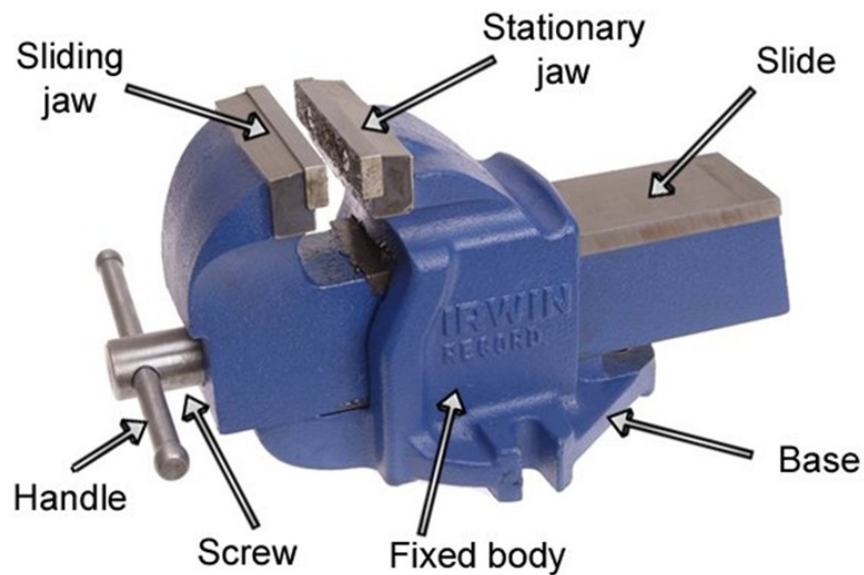
G-CLAMP



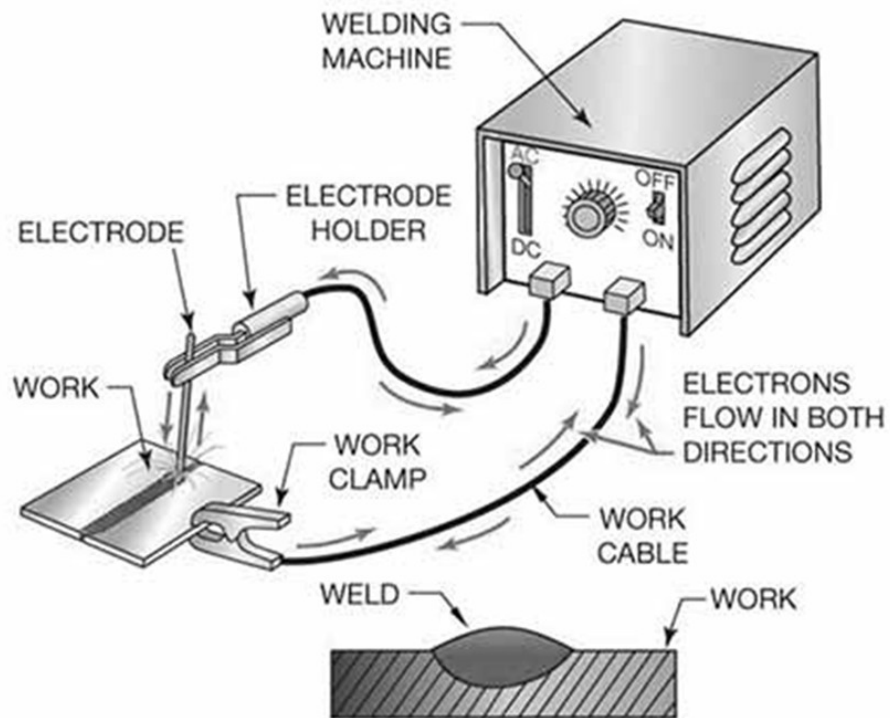
Frame:	The frame is made of steel and is the backbone of the g-clamp.
Fixed Jaw:	This is the anvil against which the workpiece is placed.
Screw:	The working piece of the G-Clamp, this will close or open the gap when adjusted.
Movable Jaw	This anvil is attached to the screw and clamps the workpiece when the screw is tightened.
Handle:	A T-handle helps tighten the screw.
Collar:	The end of the screw that houses the T-handle.



Head:	These are the ends of the spanner. You can get open and ring heads. This is a ring and open spanner. Both sides are usually the same size when it is a ring and open spanner.
Profile:	The profile of a spanner is machines so that it can accommodate the hexagonal shape of the nut or bolt
Shaft:	This portion forms the body of the spanner, and it also shows the name of the spanner.
The size of the spanner is stamped into the shaft or the head of the spanner.	



Base:	Is attached to the work table and forms the basis of the vice. Fixed Body: Houses the slide and stationary jaw.
Screw:	Moves the slide in and out when the handle is turned.
Handle:	Activates the screw.
Sliding Jaw:	This is the moving part of the vise and the size of the vice is determined by the length from side to side of the jaw.
Slide	The length of the slide determines the size of the workpiece that can be inserted.
Stationary jaw:	Does not move and the vice clamps against this jaw
Fixed Body:	Houses the slide and stationary jaw.



Welding Machine	Converts mains power to a voltage and current value suited for arc welding.
Electrode Holder:	A clamp that takes the electrode.
Work Clamp:	The earth clamp returns current to the welding machine from the work piece.

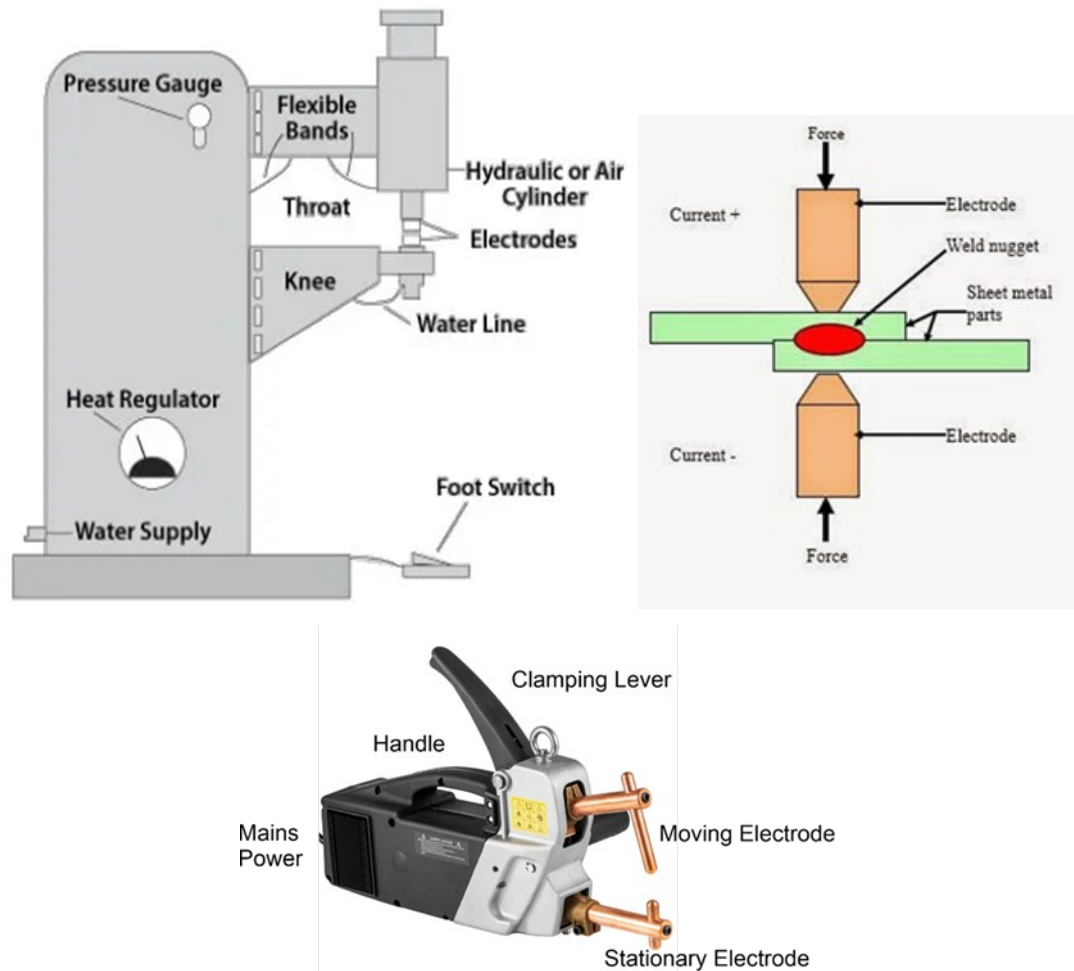
To use a welding machine safely you will need:

Leather gloves, Leather apron, Welding helmet, Tacking hammer, Clamps, Worktable, Extra electrodes, Welding machine



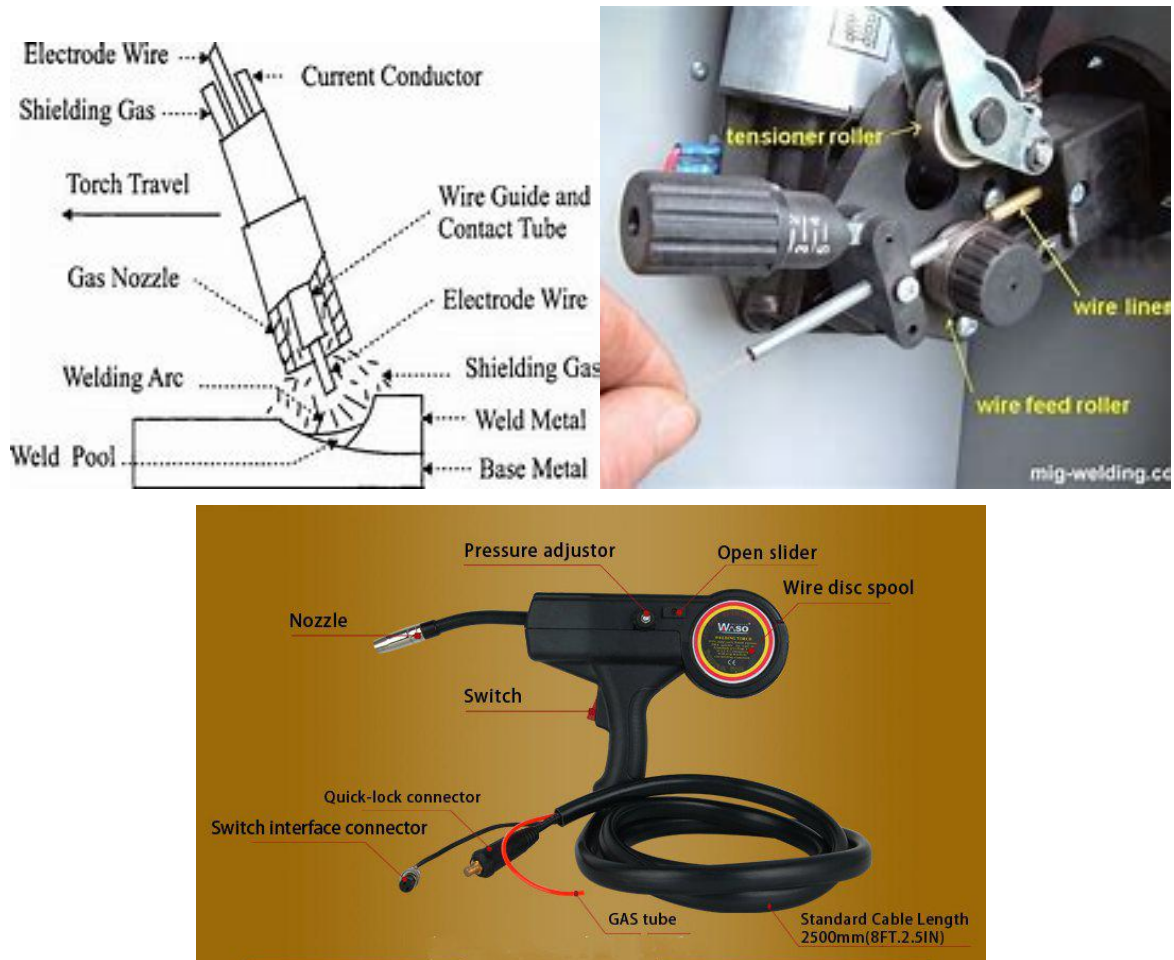
Welding machine Mig Jasic 250A - Inverter Welder.mp4

SPOT WELDER



Stationary Electrode	Usually the bottom one, it does not move.
Moving electrode:	Pushes down on the work piece and presses the metals together before welding.
Clamping Lever:	Is connected to the moving electrode and the welding switch
Handle:	used to manouvere the welder.
Mains Supply:	provides the electrical energy used to make a spot weld.

Mig 195 Welder Dual Gas No Gas 220v

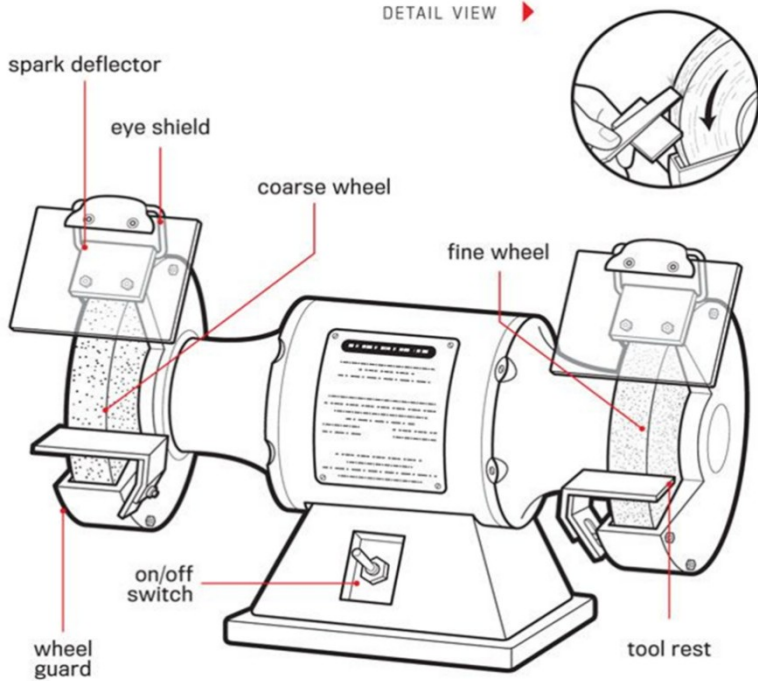


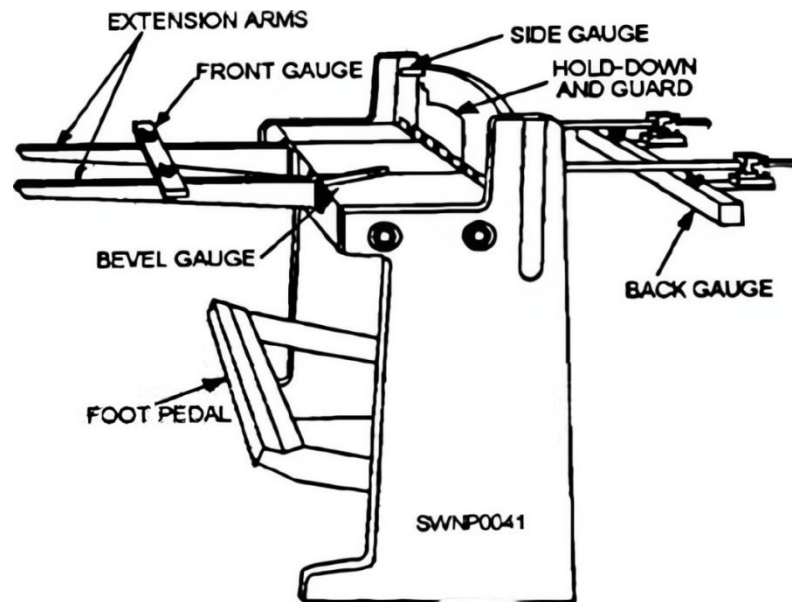
When operating a MIG welder the operator will squeeze a trigger, at which time gas will flow and the wire will be fed from the wire roll. At the same time a voltage will be applied to the wire and current will melt the wire, depositing it in the welding groove with a snapping sound



MIG_MAG Welding Machine - WTL MULTIMIG 200EX_ Unboxing and Test.mp4

BENCH GRINDER

	
On/Off Switch:	Activates the Bench Grinder
Wheel Guard:	Protects the grinder wheel against objects bumping into the grinder
Course Wheel:	Makes rough grinds and removes large portions of material but gives a rough finish
Spark Deflector and eye shield:	protects the user from flying sparks that emit from the wheel during use.
Fine wheel:	removes less material at a time, but leaves a finer more accurate cut.
Tool Rest:	For steadying the work piece.



- In order to use the shearing machine effectively, follow these instructions:
- Scribe and mark the workpiece of the to sizes and cut lines you require.
- You can now switch on the shear but take care to stay clear of the activation switch.
- Carefully insert your workpiece and align it to the blade without inserting your hands into the machine. Handle the work piece by the back end that is furthest from the cutting edge.
- Once you have aligned the piece, ensure your hands and head are free from any moving parts.
- You may now safely activate the shear.
- Remove the workpiece and repeat if needed